WORKED SOLUTION

STEP III 1987, Question 3

Pure • Arguments and Machin's Formula

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/87-s3-q3

This question has a full guided treatment in *Beyond STEP*, Section 5.1, *Machin's Formula*: https://beyondstep.keithmcnulty.org/ch05/machin.html

The key idea

Both parts rest on the polar form of a complex number. In (i), the point $(|x|, |y|)$ lies at distance $|z|$ from the origin, so $|x|\cos\alpha + |y|\sin\alpha$ is $|z|$ times a cosine. In (ii), arguments add when complex numbers multiply. So if a product of numbers like $5 - \mathrm{i}$ and $1 + \mathrm{i}$ turns out to be very nearly real, its argument gives an identity between inverse tangents. The only real care needed is the ambiguity of 2π in arguments.

Solution

Part (i)

Write $r = |z| = \sqrt{x^2 + y^2}$. If $r = 0$ then $x = y = 0$ and every term is 0, so suppose $r > 0$.

Right-hand inequality. Both sides are non-negative, so it is enough to compare squares:

$$(|x| + |y|)^2 = x^2 + y^2 + 2|x||y| \geq x^2 + y^2 = |z|^2,$$

since $|x||y| \geq 0$. Taking non-negative square roots gives $|z| \leq |x| + |y|$, with equality exactly when $x = 0$ or $y = 0$.

Left-hand inequality. The point $(|x|, |y|)$ lies in the closed first quadrant at distance r from the origin, so there is an angle $\varphi \in [0, \frac{\pi}{2}]$ with

$$|x| = r\cos\varphi, \qquad |y| = r\sin\varphi.$$

Then, by the compound-angle formula,

$$|x|\cos\alpha + |y|\sin\alpha = r(\cos\varphi\cos\alpha + \sin\varphi\sin\alpha) = r\cos(\alpha - \varphi) \leq r = |z|,$$

because $\cos(\alpha - \varphi) \leq 1$ for every real α. This proves both inequalities. Equality on the left holds when $\alpha = \varphi + 2k\pi$, that is, when $(\cos\alpha, \sin\alpha)$ points in the same direction as $(|x|, |y|)$.

(Another route to the left-hand inequality: by the Cauchy–Schwarz inequality, $|x|\cos\alpha + |y|\sin\alpha \leq \sqrt{x^2+y^2}\,\sqrt{\cos^2\alpha + \sin^2\alpha} = |z|$.)

Part (ii): the first formula

Expand step by step:

$$\begin{aligned}(5-\mathrm{i})^2 &= 25 - 10\mathrm{i} - 1 = 24 - 10\mathrm{i},\\ (5-\mathrm{i})^4 &= (24-10\mathrm{i})^2 = 576 - 480\mathrm{i} - 100 = 476 - 480\mathrm{i},\\ (5-\mathrm{i})^4(1+\mathrm{i}) &= 476 + 476\mathrm{i} - 480\mathrm{i} + 480 = 956 - 4\mathrm{i} = 4(239 - \mathrm{i}).\end{aligned}$$

Now compare arguments. Let $\theta = \tan^{-1}\frac{1}{5}$. Since $5 - \mathrm{i}$ is in the fourth quadrant, it has argument $-\theta$, and $1 + \mathrm{i}$ has argument $\frac{\pi}{4}$. Similarly $4(239 - \mathrm{i})$ has argument $-\tan^{-1}\frac{1}{239}$. Arguments add under multiplication, up to a multiple of 2π, so

$$\frac{\pi}{4} - 4\theta = -\tan^{-1}\frac{1}{239} + 2k\pi \quad \text{for some integer } k.$$

Ask yourself How do we know that $k = 0$?

We bound both sides. Since $\tan t > t$ for $0 < t < \frac{\pi}{2}$, we have $0 < \theta < \frac{1}{5}$, so $\frac{\pi}{4} - 4\theta$ lies between $\frac{\pi}{4} - \frac{4}{5} > -1$ and $\frac{\pi}{4} < 1$. Also $-\tan^{-1}\frac{1}{239}$ lies between -1 and 0. Two numbers in $(-1, 1)$ differ by less than $2 < 2\pi$, so $2k\pi = 0$ and $k = 0$. Rearranging,

$$\frac{\pi}{4} = 4\tan^{-1}\frac{1}{5} - \tan^{-1}\frac{1}{239},$$

as required.

Part (ii): the second formula

By the same reasoning, we want a product with argument

$$\frac{\pi}{4} - 3\tan^{-1}\frac{1}{4} - \tan^{-1}\frac{1}{20} - \tan^{-1}\frac{1}{1985},$$

namely $(4-\mathrm{i})^3(20-\mathrm{i})(1985-\mathrm{i})(1+\mathrm{i})$, and we hope that it is a positive real number. Leave the factor $1985 - \mathrm{i}$ until last:

$$\begin{aligned}(4-\mathrm{i})^2 &= 16 - 8\mathrm{i} - 1 = 15 - 8\mathrm{i},\\ (4-\mathrm{i})^3 &= (15-8\mathrm{i})(4-\mathrm{i}) = 60 - 15\mathrm{i} - 32\mathrm{i} - 8 = 52 - 47\mathrm{i},\\ (4-\mathrm{i})^3(20-\mathrm{i}) &= 1040 - 52\mathrm{i} - 940\mathrm{i} - 47 = 993 - 992\mathrm{i},\\ (4-\mathrm{i})^3(20-\mathrm{i})(1+\mathrm{i}) &= 993 + 993\mathrm{i} - 992\mathrm{i} + 992 = 1985 + \mathrm{i}.\end{aligned}$$

So the product so far is $1985 + \mathrm{i}$, and the last factor $1985 - \mathrm{i}$ is its conjugate: the full product is $1985^2 + 1$, which is real and positive. Equivalently, without the last factor, compare arguments:

$$\frac{\pi}{4} - 3\varphi - \psi = \tan^{-1}\frac{1}{1985} + 2k\pi, \qquad \varphi = \tan^{-1}\frac{1}{4},\ \psi = \tan^{-1}\frac{1}{20}.$$

Again $k = 0$: since $0 < \varphi < \frac{1}{4}$ and $0 < \psi < \frac{1}{20}$, the left-hand side lies between $\frac{\pi}{4} - \frac{4}{5} > -1$ and $\frac{\pi}{4} < 1$, while the right-hand side minus $2k\pi$ lies in $(0, 1)$. Rearranging,

$$\frac{\pi}{4} = 3\tan^{-1}\frac{1}{4} + \tan^{-1}\frac{1}{20} + \tan^{-1}\frac{1}{1985}.$$

Checks

- The modulus is a good check on the expansions. $|5 - \mathrm{i}|^8\,|1 + \mathrm{i}|^2 = 26^4 \cdot 2 = 913\,952$, and $956^2 + 4^2 = 913\,952$ too. Likewise $17^3 \cdot 401 \cdot 2 = 3\,940\,226 = 1985^2 + 1$.
- Numerically, $4\tan^{-1}\frac{1}{5} \approx 0.789582$, and subtracting $\tan^{-1}\frac{1}{239} \approx 0.004184$ gives $0.785398 \approx \frac{\pi}{4}$. Also $3\tan^{-1}\frac{1}{4} + \tan^{-1}\frac{1}{20} \approx 0.784894$, and adding $\tan^{-1}\frac{1}{1985} \approx 0.000504$ gives $\frac{\pi}{4}$ again.
- In (i), try $z = 3 - 4\mathrm{i}$: the bounds read $3\cos\alpha + 4\sin\alpha \le 5 \le 7$, and $3\cos\alpha + 4\sin\alpha = 5\cos(\alpha - \varphi)$ reaches 5 when $\tan\alpha = \frac{4}{3}$.

Common slips

- Writing $\arg(zw) = \arg z + \arg w$ as an exact equation and never checking the multiple of 2π. The bounds on the inverse tangents are what make the final equation true.
- Using $5 + \mathrm{i}$ in place of $5 - \mathrm{i}$: then $(5 + \mathrm{i})^4(1 + \mathrm{i}) = -4 + 956\mathrm{i}$, which is nearly imaginary rather than nearly real. The signs must be chosen so that the product is nearly real.
- In (i), forgetting that α is arbitrary, so $\cos\alpha$ or $\sin\alpha$ may be negative. The cosine form handles every α at once.

Beyond STEP

The first formula is Machin's formula (1706). Combined with the series $\tan^{-1}x = x - \frac{x^3}{3} + \frac{x^5}{5} - \cdots$, which converges fast for small x, it was used to compute π to many digits by hand. Finding such formulas is a question about factorising in the Gaussian integers $a + b\mathrm{i}$, where $1 + \mathrm{i}$, $5 \pm \mathrm{i}$ and $4 \pm \mathrm{i}$ behave like primes. This is a first taste of algebraic number theory. Davenport's *The Higher Arithmetic* discusses sums of two squares and the Gaussian integers, and Needham's *Visual Complex Analysis* shows the geometry of multiplying complex numbers.

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